

WEEK 7 - HEMATOLOGY • SAL HISTOPATHOLOGY LAB • UPDATED WITH
RELEASED REVIEW SLIDES

Blood & Bone Marrow Histopathology

Schoenlein review-slide matched • 10-slide released deck • image-first
recognition guide

Released SAL images

Official lab LOs

WBC identification

Peripheral blood vs marrow

2-page factoid

SOURCE PRIORITY: the released 10-slide SAL review deck is now the primary source. Material required by the official lab objectives but *not pictured in this review deck* is retained only in boxes labeled LO / prior-summary supplement.

Official Lab Objectives

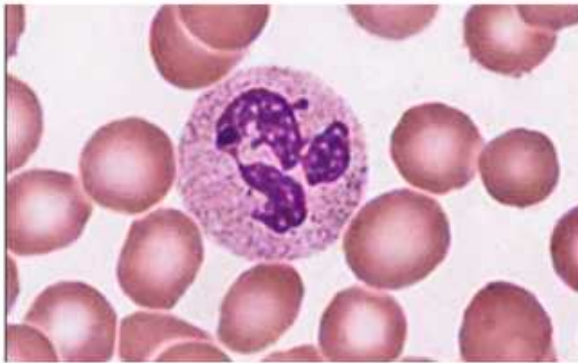
1. Recognize normal peripheral-blood RBC histology and identify anisocytosis, microcytosis, spherocytosis, sickled cells, and reticulocytosis.
2. Recognize normal histology and functions of lymphocytes, neutrophils, eosinophils, basophils, and monocytes.
3. Recognize normal peripheral-blood platelets.
4. Recognize normal bone marrow and differentiate bone marrow from peripheral blood.

WHAT THIS REVIEW DECK ACTUALLY EMPHASIZES

Identify the 5 major WBCs -> identify platelets -> recognize a normal peripheral-blood field -> recognize marrow as denser/darker with precursors -> spot megakaryocytes.

Slides 2-6: The WBC Lineup

Neutrophil

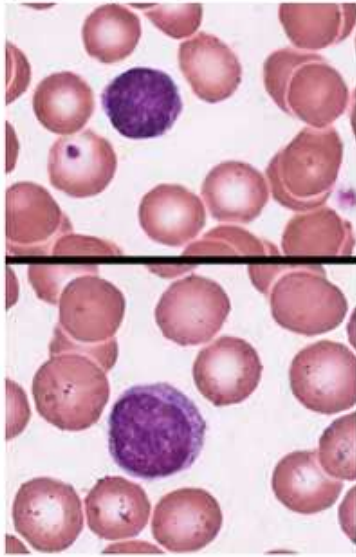


- Most abundant granulocyte
- Phagocytosis
- Lobulated nucleus

Released SAL review slide 2. The slide labels the neutrophil as the most abundant granulocyte, associates it with phagocytosis, and emphasizes its lobulated nucleus.

Slide words to know: most abundant granulocyte • phagocytosis • lobulated nucleus.

Lymphocyte



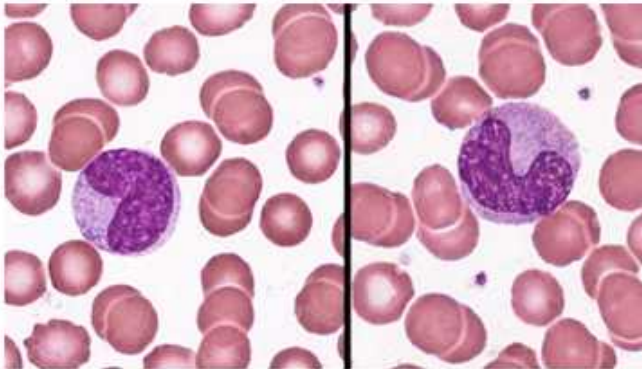
- Spherical nucleus
- Very little cytoplasm
- Cannot differentiate between T and B cells without special stain
- Agranular

Released SAL review slide 3. The slide emphasizes a spherical nucleus, very little cytoplasm, agranular morphology, and the fact that T and B cells cannot be distinguished without a special stain.

LYMPHOCYTE PRACTICAL CLUE

Round/spherical dark nucleus + very little cytoplasm. Do not call it T vs B from routine morphology.

Monocyte



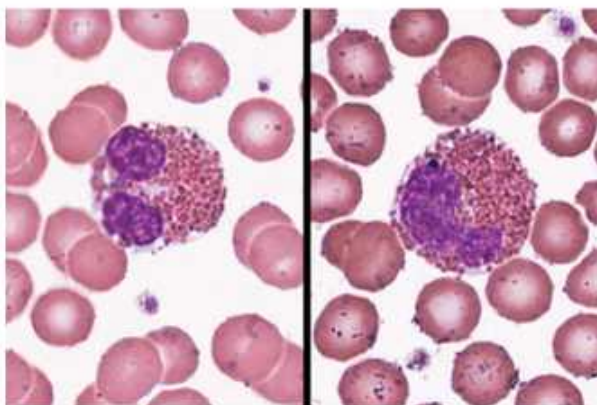
- Large, horseshoe-shaped nucleus
- Become active after entering connective tissues, where they differentiate into phagocytic cells
- Agranular

Released SAL review slide 4. The slide calls out a large horseshoe-shaped nucleus and notes that monocytes differentiate into phagocytic cells after entering connective tissue. It labels them agranular.

MONOCYTE PRACTICAL CLUE

Large cell + horseshoe/folded nucleus.

Eosinophil



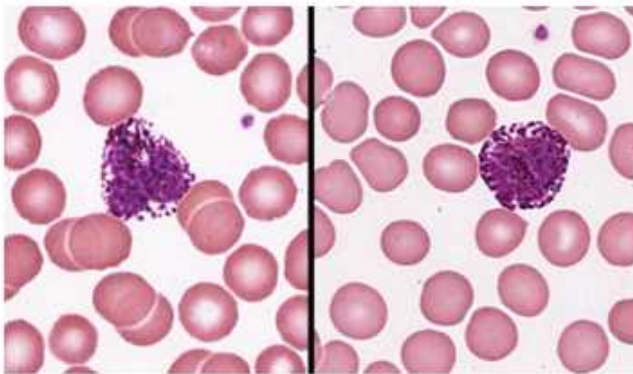
- Bright eosinophilic granules in cytoplasm
- Allergic reactions, parasitic infections, phagocytosis

Released SAL review slide 5. The review slide emphasizes bright eosinophilic cytoplasmic granules and links eosinophils to allergic reactions, parasitic infections, and phagocytosis.

EOSINOPHIL PRACTICAL CLUE

Bright pink-red/eosinophilic granules.

Basophil



-Large basophilic granules in cytoplasm

-Least numerous granulocyte

Released SAL review slide 6. The slide emphasizes large basophilic cytoplasmic granules and identifies the basophil as the least numerous granulocyte.

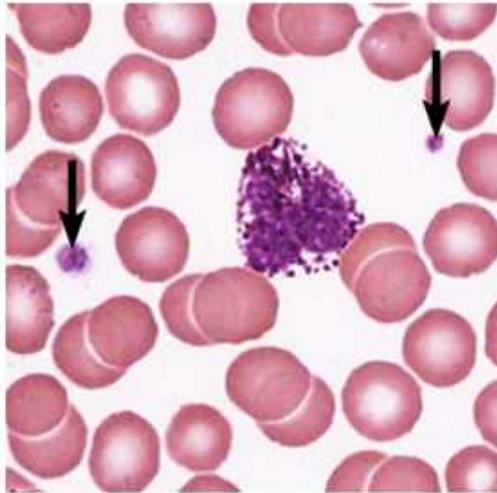
BASOPHIL PRACTICAL CLUE

Coarse dark basophilic granules dominate the cell.

Cell	Released-slide recognition cue	Released-slide function / association
Neutrophil	Lobulated nucleus	Most abundant granulocyte; phagocytosis
Lymphocyte	Spherical nucleus; very little cytoplasm; agranular	T vs B requires special stain
Monocyte	Large, horseshoe-shaped nucleus; agranular	Enters connective tissue and differentiates into phagocytic cells
Eosinophil	Bright eosinophilic granules	Allergic reactions, parasitic infections, phagocytosis
Basophil	Large basophilic granules	Least numerous granulocyte

Slide 7: Platelets

Platelets



- Small, fragmented cells located in peripheral blood
- Fragments of megakaryocytes, which are found in bone marrow
- Important functions in blood clotting and wound healing

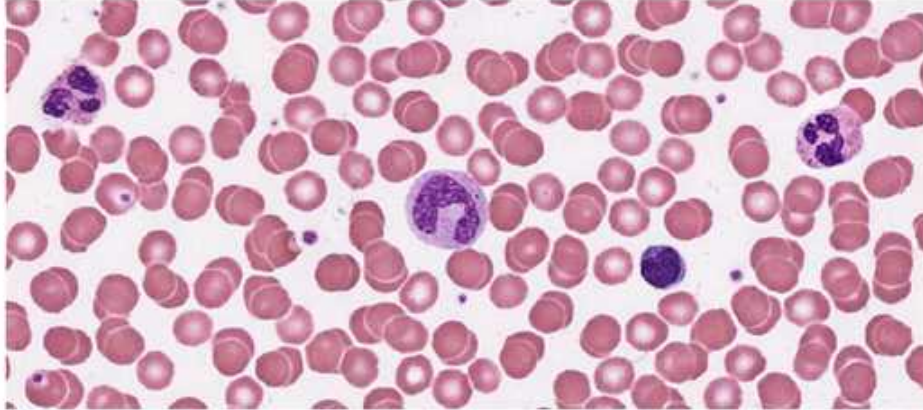
Released SAL review slide 7. The black arrows identify tiny platelet fragments in the peripheral smear. The review slide states that platelets are fragments of megakaryocytes and are important in blood clotting and wound healing.

RELEASED-SLIDE PLATELET SET

Small fragmented elements in peripheral blood -> fragments of marrow megakaryocytes -> clotting and wound healing.

Slide 8: Peripheral Blood

Peripheral Blood



Erythrocytes, Leukocytes, Platelets

Released SAL review slide 8. This whole-field image is the key normal peripheral-blood reference. The slide labels the three visible categories: erythrocytes, leukocytes, and platelets.

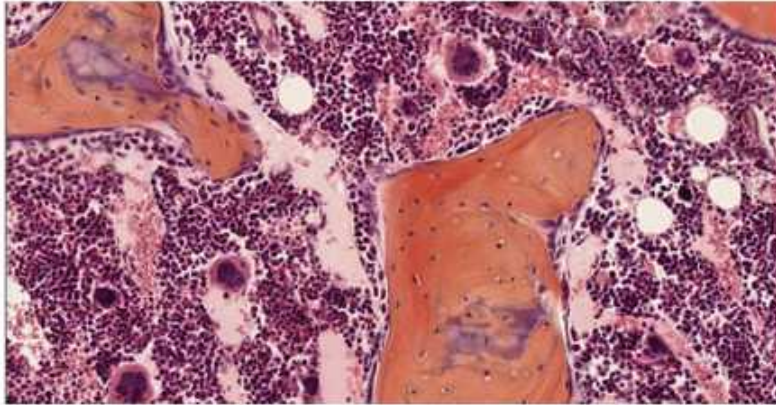
PERIPHERAL-BLOOD VISUAL GESTALT

Mostly erythrocytes, with scattered leukocytes and tiny platelets. The field is relatively open compared with marrow.

Ozzy checkpoint - click him once you can point out an erythrocyte, leukocyte, and platelet in the released field without labels.

Slides 9-10: Bone Marrow & Megakaryocytes

Bone Marrow



Denser and darker than peripheral blood; Contain precursor cells such as megakaryocytes

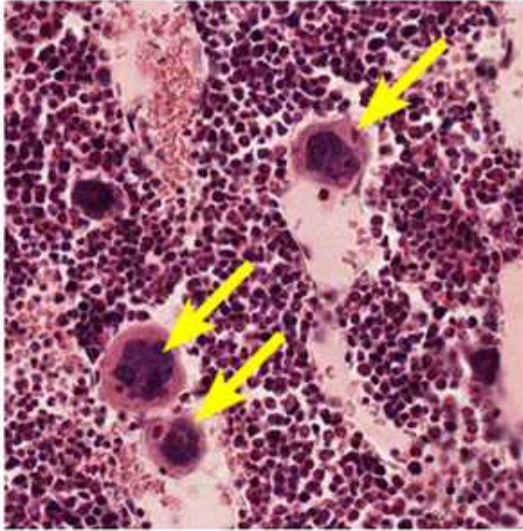
Released SAL review slide 9. The released review slide explicitly contrasts marrow with peripheral blood: marrow is denser and darker and contains precursor cells such as megakaryocytes.

DECK'S MARROW-VS-BLOOD DISTINCTION

Bone marrow = denser + darker + precursor cells.

Peripheral blood = mature erythrocytes, leukocytes, platelets in a more open field.

Megakaryocytes



- Precursors to platelets
- Large, multinucleated cells found in bone marrow

Released SAL review slide 10. Yellow arrows mark the large megakaryocytes. The review slide calls them platelet precursors and describes them as large, multinucleated cells found in bone marrow.

Keep the SAL wording: the released slide describes megakaryocytes as **large, multinucleated cells** and identifies them as platelet precursors. Other Week 7 material may use more detailed nuclear terminology, but this note preserves the review-slide wording here.

Official-LO Material Not Pictured in the Released Review Deck

The review deck does not show dedicated examples of **anisocytosis, microcytosis, spherocytosis, sickled cells, or reticulocytosis**. Those remain part of the official lab objective, so the recognition table below is retained from the earlier course summary. It is *not* presented as content from these released slides.

LO / prior-summary supplement	Recognition cue
Normal RBC	Uniform anucleate biconcave disc with central pallor roughly one-third of the diameter.
Anisocytosis	Noticeable variation in RBC size across the field.
Microcytosis	RBCs smaller than expected, often with more prominent central pallor when hypochromic.
Spherocytosis	Round dense RBCs with little or no central pallor.
Sickled cells	Crescent / elongated pointed cells.
Reticulocytosis	Polychromasia: slightly larger bluish-gray RBCs reflecting residual RNA.

WHAT NOT TO CLAIM FROM THIS DECK

The released slides also do not provide a dedicated marrow aspirate smear with staged erythroid/granulocytic precursors. They provide a marrow histology field and megakaryocyte examples. Any detailed maturation staging remains lecture/summary material rather than SAL-review-slide content.

Practical Recognition Review

LAST 60 SECONDS

Neutrophil: lobulated nucleus. **Lymphocyte:** spherical nucleus, scant cytoplasm.

Monocyte: large horseshoe nucleus. **Eosinophil:** bright eosinophilic granules.

Basophil: large basophilic granules.

Platelet: tiny megakaryocyte fragment.

Peripheral blood: mostly RBCs + scattered WBCs + tiny platelets.

Marrow: denser/darker + precursors; **megakaryocyte** is the giant platelet precursor.

Updated source hierarchy: Schoenlein Blood and Bone Marrow Histopathology Lab student review slides are primary. Official-LO RBC-abnormality material not pictured in the released deck is retained only as a labeled prior-summary supplement.